Smart Bed: Use Cases

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What a use case is

A use case describes one specific way a real person interacts with the system to achieve a goal: who the actor is, what triggers the interaction, the main steps, what can go wrong, and what counts as success. Each use case below follows the same structure: actor, goal, trigger, preconditions, main flow, exceptions, and outcome.

Support tiers used across these use cases

Tier	Score	Meaning	What the smart bed does	Who is alerted
1: Light support	0 to 3	Can get up alone but finds it hard	Warms the room before wake-up, gentle cue, logs time in bed	The user only
2: Prompted	4 to 6	Safe alone but benefits from prompts	Bed-exit alert if not back in time, darker-hours threshold, climate alerts	User, plus family if opted in
3: Assisted	7 to 10	Needs help or real fall risk	Immediate bed-exit alert, multi-modal alerts, night escalation	Named carer, or care staff if in a care home
4: Continuous	11 to 15	Cannot leave bed unaided	Constant monitoring, escalating alerts, long-stay-in-bed flag	Care staff or call centre

Mock scenarios (personas behind the use cases)

- **Aoife, 34, Tier 1 (score 3).** Diagnosed depression and some anxiety; mornings are hardest. No physical or visual impairment. The heater warms her room before her chosen time and the LED glows as a gentle cue. Data is hers alone unless she chooses to share it.
- **Niamh, 45, Tier 1 (score 3).** Blind since birth, otherwise healthy. Her low score shows a light tier is not always about age — every alert for her must be audible, since the LED is useless to her.
- **Tom, 68, Tier 2 (score 6).** Lives alone, early memory changes, occasional low mood, some stiffness and mild vision loss. If the bed stays empty 20 minutes during dark hours, the buzzer sounds and his daughter is alerted.
- **Maureen, 79, Tier 3 (score 7).** Arthritis, walking frame, macular degeneration (low vision), occasional low mood. Every alert is both visual and audible. Used as the worked example for the full use case below.
- **Seán, 86, Tier 4 (score 11).** In a care home after a stroke, needs two staff to get up, depression and mild cognitive change. Staff see occupancy and climate on a dashboard; an exit raises an immediate alert.

Use case: Night-time bed exit and safety alert

Actor: Maureen, 79 — Tier 3 (Assisted). Arthritis, uses a walking frame, macular degeneration (low vision), occasional low mood.

Secondary actor: Care staff (primary responder, since Tier 3–4 residents are in a care setting), with her daughter informed afterward as next of kin.

Goal: Maureen gets in and out of bed safely overnight, and staff are alerted in time if something seems wrong.

Trigger: Maureen gets into bed for the night, or gets out of bed unexpectedly.

Preconditions:

- The system is installed and wired (DHT22, pressure mat, LDR, MCP3008, status LED, buzzer, heater relay)
- Maureen's tier (3) and thresholds are configured
- The care staff dashboard is set as her primary alert route, and her daughter is recorded as next of kin with consent given

Main flow:

1. Maureen gets into bed. The pressure mat detects occupancy.
2. The DHT22 reads room temperature. If it's below the low threshold, the heater switches on and the status LED lights.
3. The LDR detects it's dark, so the system switches to night-time alert rules (shorter grace period, louder buzzer).
4. Maureen gets up during the night. The pressure mat detects the bed is empty.
5. A 20-minute timer starts (her Tier 3 grace period).
6. If she hasn't returned within 20 minutes, the local buzzer sounds and the care staff dashboard is alerted immediately.
7. Staff check on her. Once resolved, her daughter is informed as a courtesy update.
8. If temperature or humidity crosses a safe limit at any point, the heater cuts off and a separate buzzer pattern sounds, regardless of occupancy state.

Exceptions:

- Maureen returns to bed within 20 minutes: no alert is sent; the event is just logged.
- The pressure mat gives a false ‘empty’ reading (e.g. she's sitting on the edge of the bed): a short buffer should run before the timer starts, to avoid nuisance alerts.
- A sensor fails or gives no reading: the system logs an error rather than crashing, and flags that it needs attention.
- Because Maureen has low vision, every alert must be audible as well as visual.

Outcome: Either Maureen safely returns to bed and nothing further happens, or staff are alerted in time to check on her — without requiring her to wear a device, press a button, or do anything other than get in and out of bed as normal.

Checklist derived from this use case

A tickable setup checklist built directly from the use case above, for a Tier 3 resident in a care setting.

- Score the user against the tier table (age, mood, mental health, physical impairment, visual impairment)
- Confirm tier and support level (Tier 3: Assisted)
- Install the pressure mat under the mattress for occupancy
- Install the DHT22 for temperature and humidity

- Install the LDR for ambient light
- Wire the pressure mat and LDR through the MCP3008 ADC
- Wire the status LED to mirror the heater's on/off state
- Wire the buzzer for alerts
- Wire the heater through a relay (never direct GPIO)
- Set temperature low and high thresholds
- Set the humidity threshold
- Set the night-time bed-exit grace period (e.g. 20 minutes for Tier 3)
- Assign a distinct buzzer pattern per alert type (bed-exit, overheat, humidity)
- Confirm all alerts are audible, not only visual, for low-vision users
- Set the primary alert route to care staff (locked for Tier 3–4) and add next of kin to inform
- Test the bed-exit alert manually
- Test the overheat alert manually
- Test the humidity alert manually
- Confirm the care staff dashboard and the next-of-kin contact both receive the correct notifications
- Walk the user through what each sound and light means
- Monitor logs daily for the first week for false alarms
- Adjust thresholds or grace period if nuisance alerts occur
- Re-score the user's tier monthly or after any health change